Dataset & Business Context

| A 1994 Census dataset used for credit-risk screening — data governance starts at row zero.

UCI Adult Census Income Dataset

- **Source:** Kaggle / UCI ML Repository (CC0 Public Domain)
- **Records: 32,561** rows × **15** columns
- **Target:** Binary — income >50K vs ≤50K
- **Era:** 1994 US Census (reflects historical inequalities)

Business Scenario

- **Application**: Automated credit scoring system.
- **Social Impact**: Incorrect predictions lead to **Financial Exclusion**.
- **Governance Mandate**: Fairness is not a "feature"—it is a regulatory and ethical requirement.

Data Governance Framing

- **GDPR Art. 22**: right to explanation for automated decisions
- **PDPA (Singapore)**: purpose limitation & consent
- **Our approach**: ML models as evidence of data quality, not the end goal — a data-centric mindset

32,561
Total Records

6+8 -> 1
Num+Cat -> Target

75 : 25
Class Ratio

5.6%
Missing Values

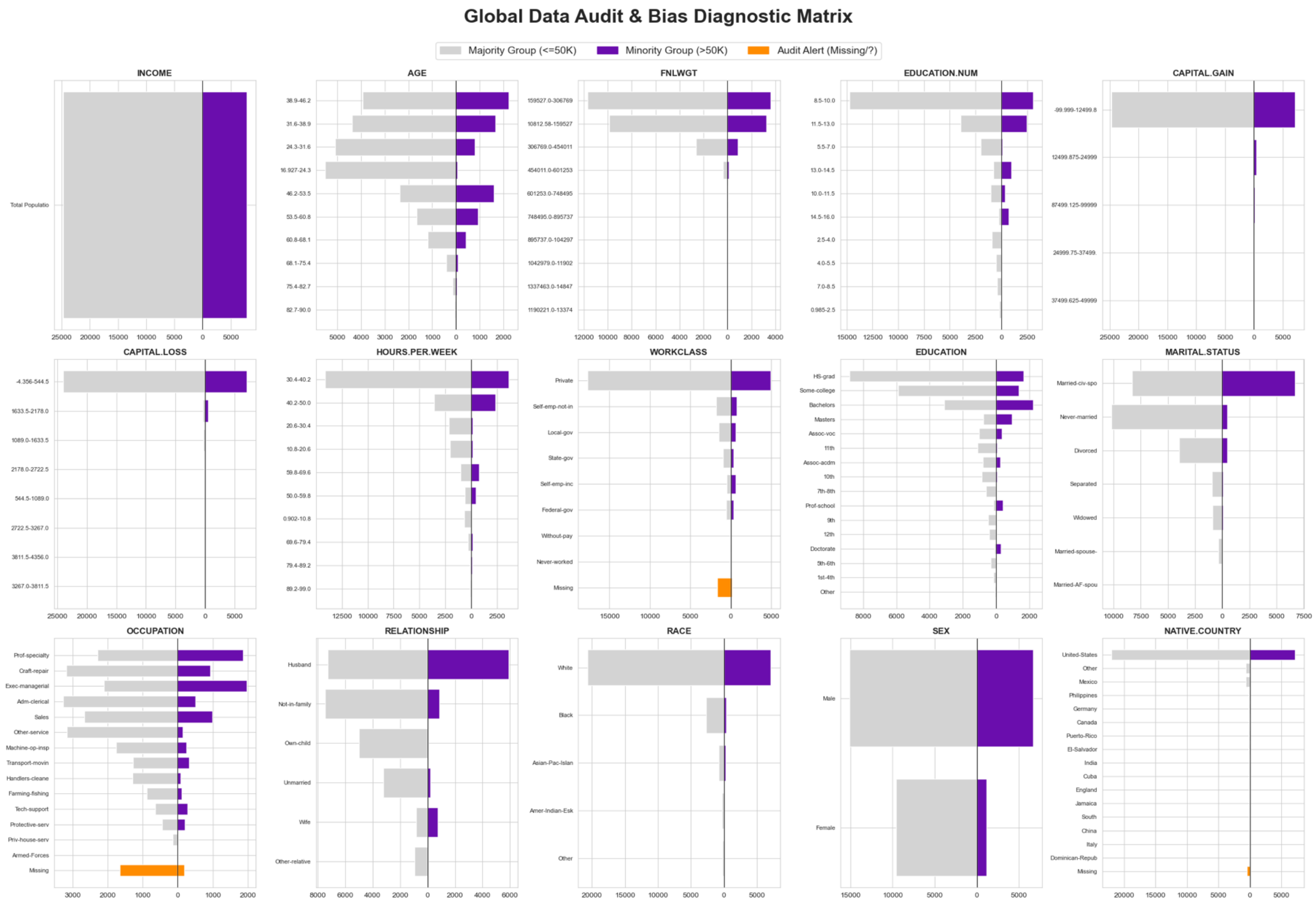
24
24 duplicate rows

No.	Feature	Dtype	Audit Label / Risk Focus
1	Age	Num	Seniority / Experience
2	Workclass	Cat	Employment Sector
3	fnlwgt	Num	Sampling Scale (Noise?)
4	Education	Cat	Skill Level (Text)
5	Education.num	Num	Core Skill (Merit)
6	Marital.status	Cat	Identity Proxy?
7	Occupation	Cat	Job Type / Role
8	Relationship	Cat	Identity Proxy?
9	Race	Cat	Protected (Legal)
10	Sex	Cat	Protected (Legal)
11	Capital.gain	Num	Wealth Indicator
12	Capital.loss	Num	Wealth Indicator
13	Hours.per.week	Num	Effort / Workload
14	Native.country	Cat	Origin Bias
-	**Income (Target)**	**Binary**	**Class Imbalance**

Data Quality Audit | Global Distribution Scan: Identifying Data Skews and Governance Red Flags

This diagnostic audit identifies structural skews and governance risks within the data distributions.

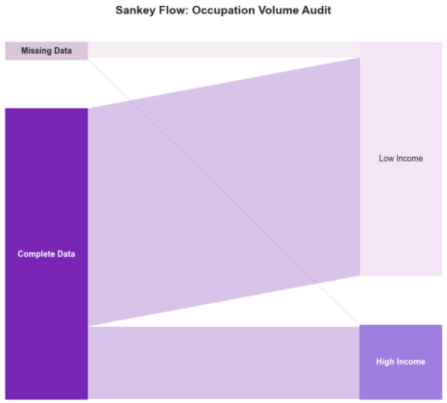
Red Flags	Related Features	The "Clue" (Observation)	Potential Risk
Target	*Income*	Class Skew (75:25)	Majority Bias
Integrity	*Occupation, Workclass*	Systemic "?" Blocks	MNAR / Info Loss
Redundancy	*Rel., Marital, Sex, Edu.*	Semantic Convergence	Proxy Leakage
Sparsity	*Race, Native-country*	High Cardinality / Skew	Generalization Risk
Numerical	*Capital.gain, Loss*	Zero-Inflation	Outlier Sensitivity
Scaling	*fnlwgt*	Magnitude Disparity	Weight Distortion



Integrity & Redundancy Audit | Identifying MNAR Signals and Multicollinearity

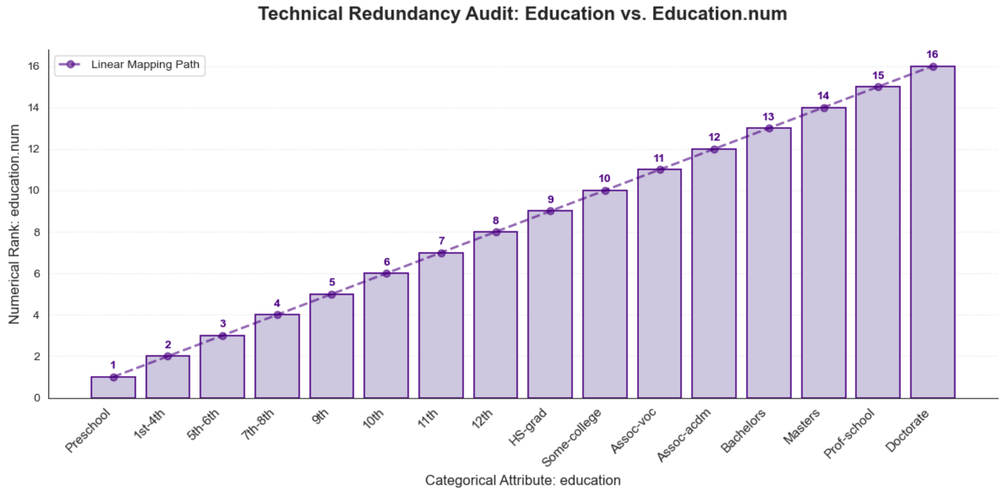
Missing Data Audit: MNAR Signal

- **Co-occurrence Pattern**: *workclass* and *occupation* missingness (~5.6%) is highly synchronized, indicating a systematic rather than random pattern.
- **MNAR Evidence**: High-income probability for missing records is <50% of the baseline; "data silence" serves as a strong **negative proxy** for income.
- **Final Strategy**: Rejected **mode-based** imputation to prevent signal loss. Implemented **"Unknown" encoding** (via SimpleImputer) to treat missingness as a distinct, informative feature.



Education Audit: Redundancy Check

- **Identical Information**: The categorical *education* and numerical *education.num* share a **1:1 perfect mapping** (Correlation = 1.0).
- **The Problem:** Keeping both introduces **perfect multicollinearity**, which destabilizes model coefficients and artificially inflates feature importance.
- **The Decision**: **Ablate** the categorical string; **Retain** the numerical rank to leverage ordinal relationships and reduce dimensionality.



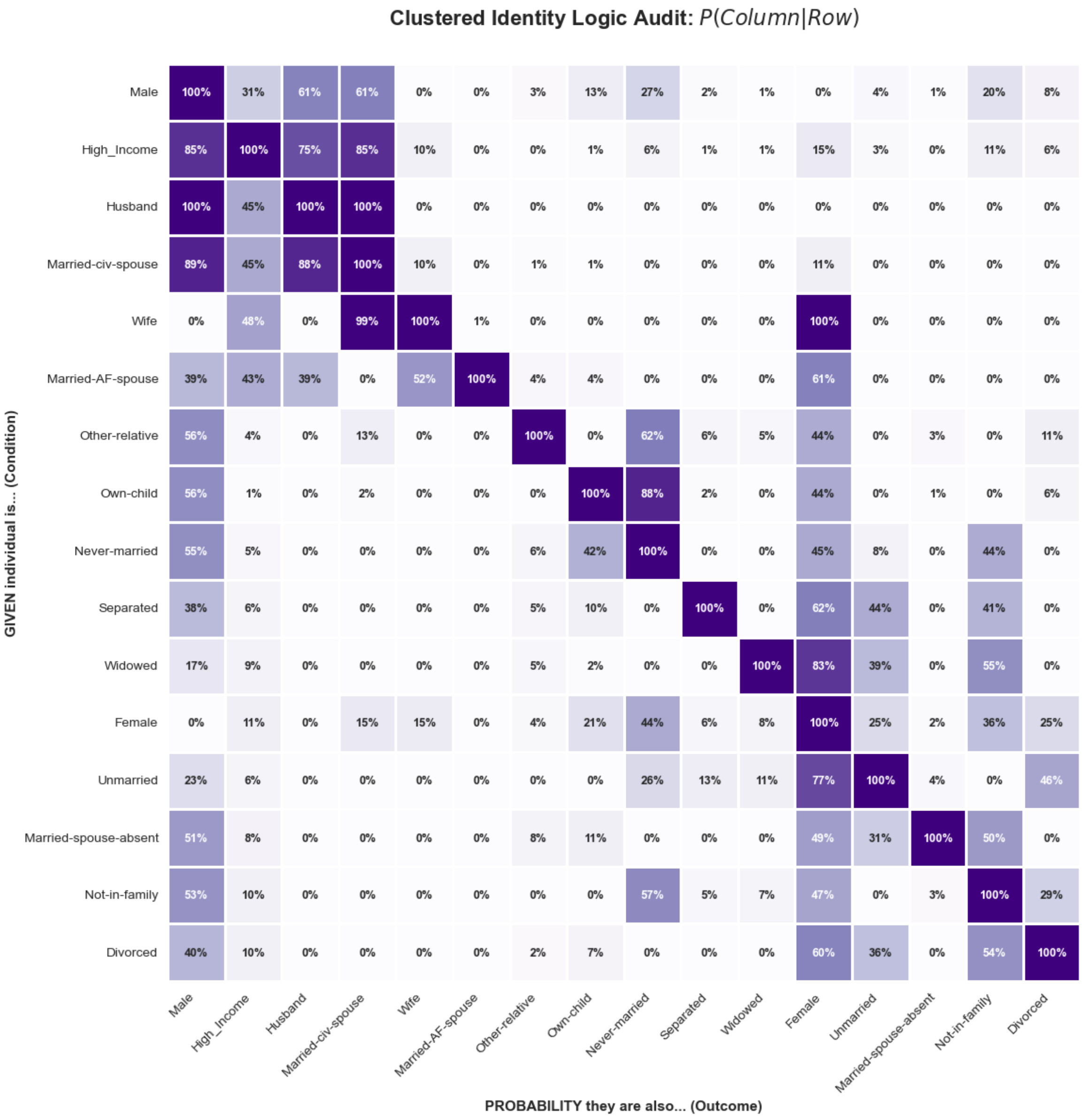
Logic Audit & Proxy Detection | Uncovering Information Redundancy and Entangled Identity Markers

Some features are not independent; they are just **echoes** of each other. These features are "double-counting" the same identity information.

Relationships	***Implied Sex***	***Implied Marital Status***	**Audit Insight**
Husband	100% Male	100% Married	Total Proxy (Redundant)
Wife	100% Female	100% Married	Total Proxy (Redundant)
Unmarried	Mixed	100% Single/Divorced	Status Redundancy
Divorced*	Mixed	100% Divorced	Logical Overlap
Own-child	Mixed	90%+ Never-married	Strong Correlation
Other-relative	Mixed	Mixed	Weak Signal

The Risk: Identity Shortcuts

- **Confusing the Model**: This "tangle" makes it hard for the model to see what truly matters (like skills or job type).
- **Bias Warning**: The model might take a **shortcut** by looking at a person's social role instead of their professional merit.



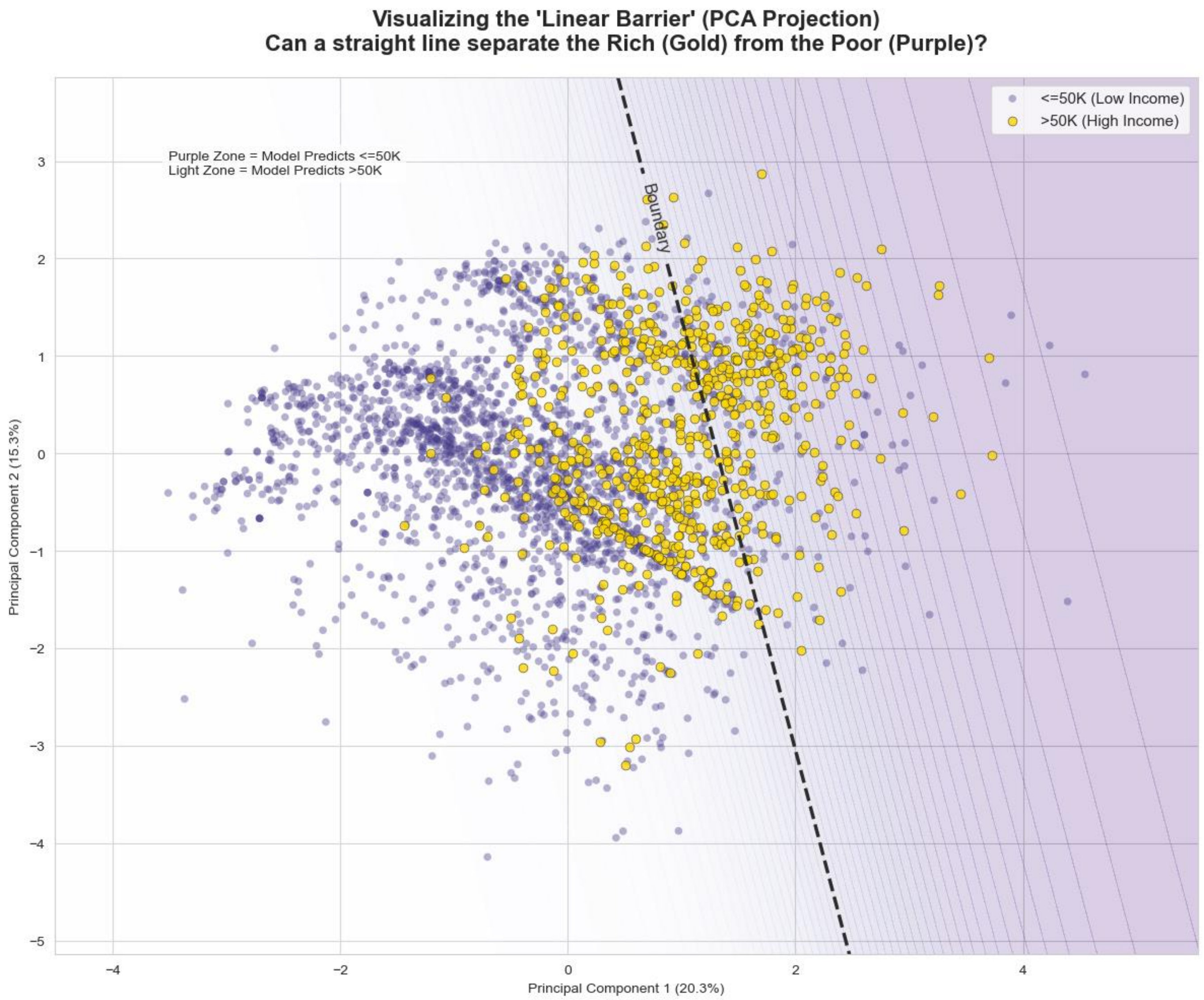
The Accuracy Paradox Diagnostic | Visualizing why Model Performance is Deceptive

Model-Aided Audit: 3,000-Sample

Group / Class	Precision	Recall	F1-score	Status
Low Income (<=50K)	86.9%	92.1%	89.4%	✅ Good
High Income (>50K)	68.4%	55.4%	61.2%	⚠ FAIL
Overall Model	-	83.4% Accuracy		Misleading

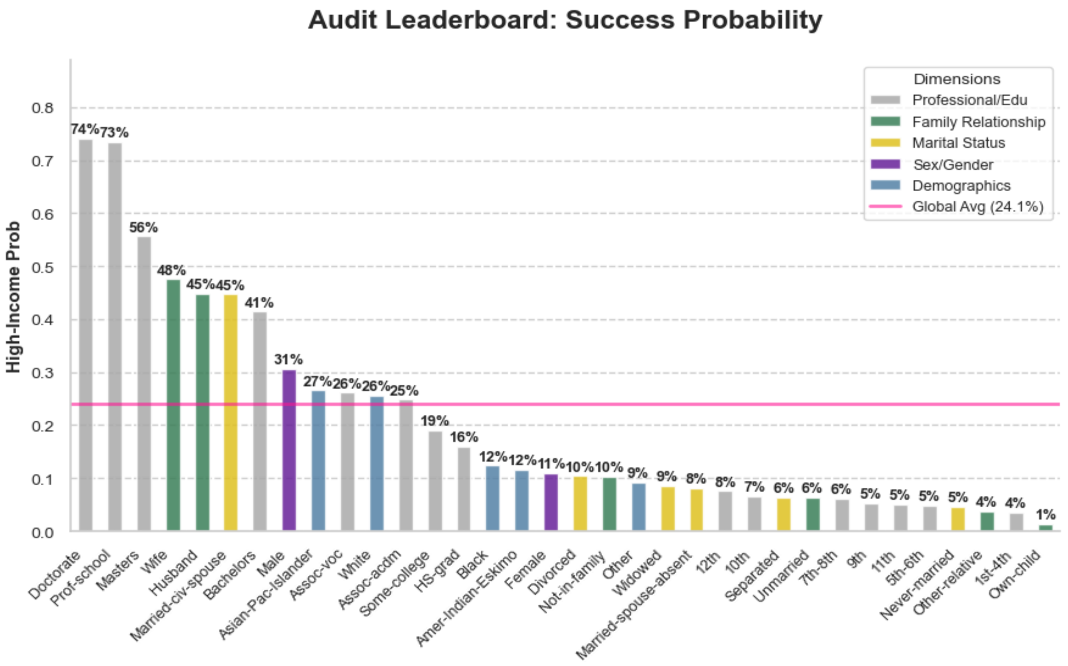
Group	FNR	Accuracy	Hidden Problem
Male	43.2%	79.3%	—
Female	51.2%	91.4%	Accuracy masks bias
Gap	+8.0pp	+12.1pp	Paradox!

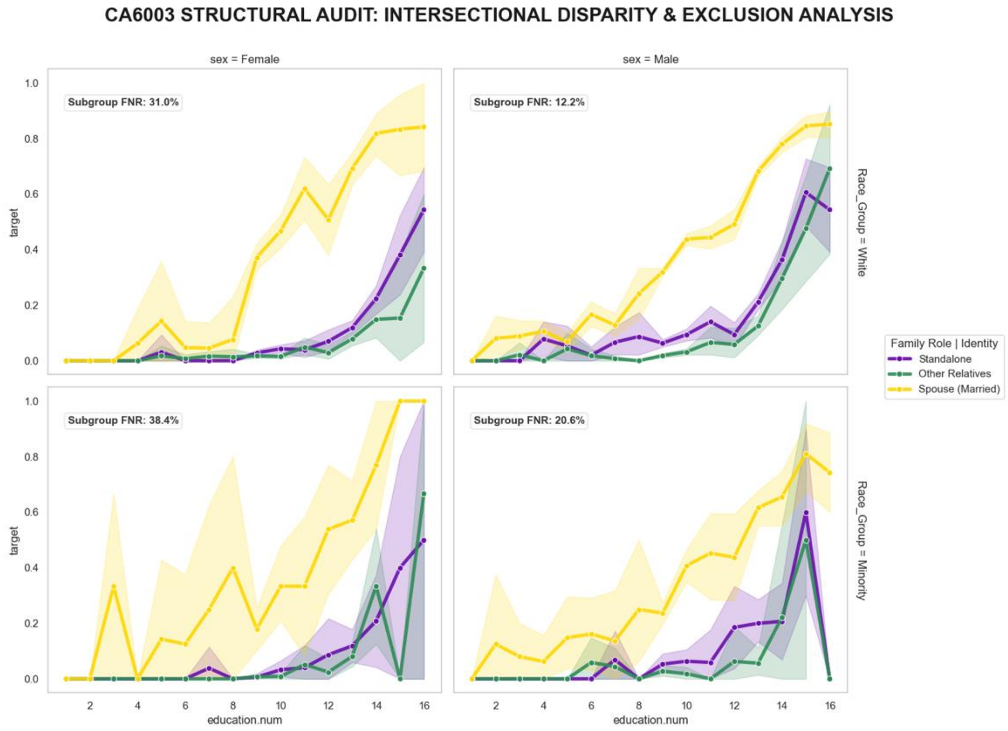
- **Global Summary**: 83% Accuracy = **"Lazy Fitting"**
- **Class High Income**: **55% Recall** (Failing)
- **Demographic Evidence**: **Female FNR > 50%** (Identity Shortcut)
- **PCA Insight**: **Majority-Biased Boundary** (Sacrificing the Minority)



Structural Bias Diagnosis | Auditing Intersectional Failure & Systemic Exclusion

- **Historical Proxy for Success:** The model adopts "Marriage" as a predictive shortcut, reflecting 1990s socioeconomic biases rather than true merit.
- **The Intersectional Glass Ceiling:** Education fails to bridge the gap for marginalized groups; single minority females face a "suppressed" success curve regardless of skill.
- **Systemic Exclusion (The 9.74x Gap):** While the reference group is safe, at-risk groups face a **9.74x higher risk** of being incorrectly rejected (78.2% FNR).





Intersectional Profile	Sample Size	FNR	Exclusion Multiplier
Female | Single | Minority	1,771	78.2%	**9.74x**
Female | Single | White	7,124	74.2%	9.24x
Male | Single | Minority	1,280	69.3%	8.63x
Male | Single | White	6,969	48.7%	6.07x
Female | Married | Minority	3,581	15.6%	1.95x
Male | Married | Minority	1,336	13.3%	1.65x
Female | Married | White	1,518	8.4%	1.05x
Male | Married | White	12,205	8.0%	1.00x

Strategic Feature Engineering | From Data Insights to Model Design

Defining the Bounds: Naive vs. Robust To isolate the impact of feature engineering, we established two contrasting baselines:

- **The "Raw" Pipeline (Naive Baseline)**
    - **Philosophy:** "Minimum Viable Product." Focuses strictly on **Technical Feasibility** (making code run).
    - **Role:** Ignores data defects to establish the performance **floor**.

The "Optimized" Pipeline (Robust Standard)

- **Philosophy:** "Best Practice." Focuses on **Statistical Validity**.
- **Role:** Systematically corrects defects (e.g., MNAR, Skewness) to maximize **signal quality**.

From Insight to Architecture This log acts as the strategic bridge between EDA and Modeling. It documents how we translated qualitative insights (e.g., "data is skewed") into quantitative pipeline decisions, ensuring every step is **evidence-based**.

Dimension	Raw Pipeline	Optimized Pipeline	The "Logic" (Why it matters)
Duplicate Rows	Removed	Removed	Data Integrity: Eliminates potential leakage between training and testing sets.
Missing Values	Impute Mode	'Unknown' Category	Activates MNAR Signal: Preserves "data silence" as a predictive feature for income.
Data Skewness	Original Scale	Yeo-Johnson Transform	Reduces Wealth Bias: Normalizes extreme outliers in Capital Gain/Loss.
Feature Scaling	No Scaling	Standard Scaler	Leveling the Field: Prevents large-scale features from dominating model weights.
Redundancy	Kept All Features	Feature Selection	Complexity Control: Drops noise (fnlwgt) and redundant text (Education).
Categorical OHE	Standard OHE	Standard OHE	Technical Feasibility: Ensures the model can mathematically process text inputs.
Class Imbalance	None (Default)	None (Controlled)	We deliberately keep weights neutral before introducing bias mitigation strategies.

The Experiment Architecture | A Modular Pipeline Framework

1. SPLIT
train_test_split
stratify=income
80/20

2. IMPUTE
A.Mode /
B.Constant='Unknown'
(train only)

3.ENCODE
A.Standard OHE/
B.Group Rare Categories

4. TRANSFORM
A.None / B. Yeo-Johnson /
C.**Polynomial**

5. SCALE
A.None /
B.StandardScaler
(train only)

6. MODEL
LogisticRegression
/RandomForest(Ref)

Optimization Strategy

- We replaced the "black box" with a transparent, incremental pipeline.
- **Decoupled Streams:** Categorical (MNAR) and Numerical features are optimized independently.
- **Progressive Complexity:** We systematically layer techniques—**Scaling (P3) → Distribution (P4) → Non-Linearity (P5)**—to pinpoint the precise source of accuracy gains.

Critical Execution Order

- Step 1 → 2/3 (Leakage Firewall): Split First, Learn Later.
- Step 2 → 3 (Dependency): Fill First, Calculate Later.
- Step 4 → 5 (Math Logic): Shape First, Size Later.

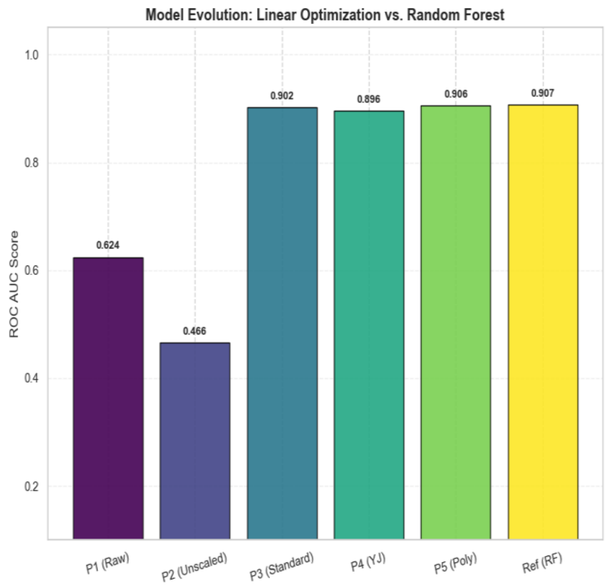
Pipeline	Step 2: Imputation	Step 3: Encode (Cat)	Step 4: Transform (Num)	Step 5: Scale	Step 6: Model	The "Logic" (Experimental Role)
P1 (Raw)	Mean / Mode	Basic OHE	None	None	Logistic Regression	Technical Floor: Minimal processing. Ignores MNAR and skewness to establish the worst-case baseline.
P2 (Unscaled)	Median / 'Unknown'	Adv. OHE + Group Rare	None	Missing	Logistic Regression	Quality Benchmark: Clean inputs (MNAR handled) but lacks mathematical scaling. Tests if clean data alone is enough.
P3 (Standard)	Median / 'Unknown'	Adv. OHE + Group Rare	None	StandardScaler	Logistic Regression	Linear Baseline: The industry standard. Applies Z-score scaling to the cleaned data for stable convergence.
P4 (Optimized A)	Median / 'Unknown'	Adv. OHE + Group Rare	Yeo-Johnson	StandardScaler	Logistic Regression	Distribution Fix: Hypothesis that normalizing skewed data (fixing Gaussian assumptions) improves linear performance.
P5 (Optimized B)	Median / 'Unknown'	Adv. OHE + Group Rare	Polynomial (Deg=2)	StandardScaler	Logistic Regression	Non-Linearity: Adds complexity (interaction terms) to the linear model to capture curved relationships (e.g., Age).
Ref (nonlinearity)	Median / 'Unknown'	Adv. OHE + Group Rare	None	None (Not Required)	Random Forest	Performance Ceiling: A complex non-linear model used as the "target to beat" for our engineered linear models.

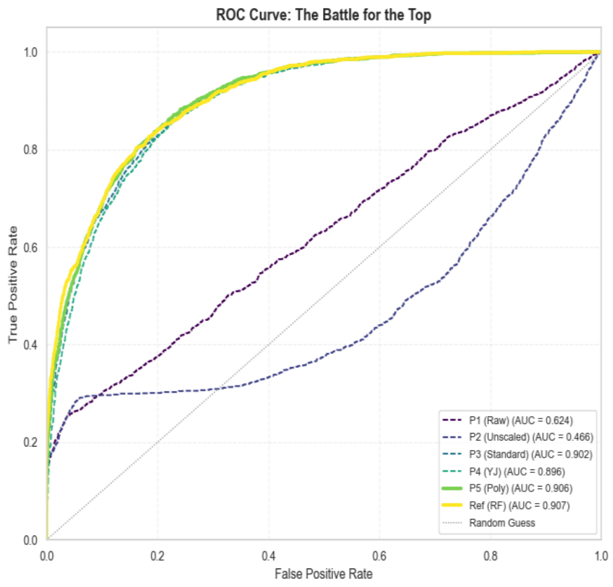
Experimental Results | Bridging the Gap: Linear vs. Non-Linear Benchmarks

Results Summary:
Feature Engineering Bridges the Gap

We successfully transformed a failing linear model (AUC 0.46) into a high-performance contender (AUC 0.906) that matches the Random Forest benchmark, proving that rigorous preprocessing is as critical as model selection.

Experiment ID	Pipeline Description	AUC (Main Metric)	F1-Score	Recall	FNR (Miss Rate)
P1	Raw Baseline (Naive)	0.624	0.358	0.239	76.1%
P2	Unscaled (Control)	0.466 ⚠️	0.365	0.249	75.1%
P3	Standard (Linear Base)	0.902	0.659	0.598	40.2%
P4	Yeo-Johnson (Dist. Opt)	0.896	0.644	0.589	41.1%
P5	Polynomial (Non-Linear)	0.906 🏆	0.663	0.599	40.1%
Ref	Random Forest (Benchmark)	0.907	0.652	0.554	44.6%





- **A. The "Scaling Trap" (P2 vs P3) P2 Collapsed (0.46).** Proves StandardScaler is mandatory. Without it, the solver **failed to converge** on high-magnitude features.
- **B. The "Distribution" Reality (P3 vs P4) Hypothesis Rejected.** Yeo-Johnson failed. The **zero-inflated** nature (92% zeros) introduced noise rather than signal.
- **C. The "Non-Linear" Victory (P3 vs P5) Hypothesis Confirmed (🏆).** Polynomial features captured **curved relationships** (e.g., Age), boosting AUC to **0.906**.
- **D. The "Engineering" ROI (P5 vs Ref) Gap Closed.** Tied with Random Forest (0.907). We achieved **"Black-Box Performance"** with **"White-Box Interpretability."**

Bias & Fairness Analysis | Identifying Domestic Status Bias

Feature	Group	Sample Size (n)	Recall (TPR) ↑	False Positive Rate (FPR) ↓
Gender	Male	4335	60.9%	9.7%
	Female	2173	54.4%	1.8%
Race	White	5581	60.4%	7.1%
	Black	632	54.2%	2.0%
	Asian-Pac-Islander	199	56.3%	12.6%
	Amer-Indian-Eskimo	53	42.9%	4.3%
	Other	43	66.7%	5.0%
Relationship	Husband	2618	65.4%	19.5%
	Wife	326	67.3%	16.5%
	Not-in-family	1658	33.1%	0.8%
	Unmarried	704	27.3%	0.3%
	Own-child	1000	14.3%	0.1%
	Other-relative	202	20.0%	0.0%
Marital Status	Married-civ-spouse	2995	65.3%	18.7%
	Never-married	2075	27.3%	0.4%
	Divorced	907	32.0%	0.5%
	Widowed	215	25.0%	0.5%
	Separated	211	23.1%	0.5%
	Married-spouse-absent	101	60.0%	1.1%
	Married-AF-spouse	4	0.0%	0.0%

Core Issue: The "Marriage Trap"

The model still relies on a **domestic shortcut**, using marital and household roles as proxies for wealth. As a result, it **prioritizes family status over individual attributes**. This causes two systemic errors:

- **Over-estimation:** Married groups receive an unearned predictive advantage, with much higher false positive rates.
- **Recognition gap:** High earners outside traditional household roles (e.g., never-married or own-child) are systematically under-recognized.

Mitigation Strategies

Scheme	Strategy	Governance Logic (Key Mechanism)
A	**Label Balancing**	**Fix Distribution:** Uses class_weight to solve the *Global Imbalance* (Rare high-earners).
B	**Sample Re-weighting**	**Promote Equity:** Uses subgroup weights to penalize *Local Bias* (*The Marriage Trap*).
C	Hybrid Mitigation	**Synergize A & B:** Simultaneously applies both weights to **Maximize Inclusion** (Best Recall).

Bias & Fairness Analysis | After Mitigation

Strategic Trade-off (Performance vs. Fairness)

Metric	Baseline (P5)	Scheme 1 & 3 (Hybrid)	Scheme 2 (Pure Reweight)
Strategy	Default Status Quo	Maximize Inclusion	Strict Fairness
Recall (TPR)	59.4%	83.1% (Best)	59.9% (Drop)
Precision	74.0%	57.7% (Cost)	74.2% (Best)
Gender Gap	~10%	~12%	< 2% (Solved)
Policy Verdict	❌ Neglects Talent	✅ Safety Net (Inclusion)	⚠️ Meritocracy (Precision)

Scheme 1 & 3 is the only viable option for a "Safety Net" policy, as it ensures no high-potential individual is left behind (83% Recall), despite the cost to precision.

Structural Bias Audit (The "Marriage Trap")

Subgroup (FPR)	Baseline	Scheme 1 & 3 (Hybrid)	Scheme 2 (Pure Reweight)
Husband	19.5%	52.7% 🚩 (Severe Bias)	19.5% (Normalized)
Wife	14.1%	51.8% 🚩	16.5% (Normalized)
Unmarried	0.3%	1.5%	0.3%
Mechanism	Relies on Marriage Proxy	Amplifies Proxy to boost Recall	Penalizes Proxy to fix Gap

To achieve high recall (Scheme 1 & 3), the model aggressively uses "Marriage" as a shortcut. Scheme 2 successfully removes this shortcut, but at the cost of missing valid candidates.

Conclusions | The Cost of Fairness & Strategic Verdict

1 Structural Bias

- **"Hiding columns is not enough."** Removing 'Sex' failed because the model found a **"sociological backdoor"**—Marriage. True governance must target these hidden proxies, not just protected attributes.

2 The Fairness Cost

- **"Strict fairness kills visibility."** Our "Strict Fairness" model (Scheme 2) successfully closed the gender gap (<2%), but caused a **40% collapse in Recall**. We cannot have perfect equality without sacrificing performance.

3 Strategic Inclusion

- **"Inclusion requires tolerance."** To build a **"Safety Net"** (83% Recall) that leaves no talent behind, we must accept higher False Positives for privileged groups (Husbands). This is a necessary trade-off for social impact.

Project Summary

Dataset: 32,561 records × 15 features (1994 Census)
Task: Binary income prediction for credit approval

Key Finding: The "Marriage Trap" — Model relies on marital status as a primary predictor.

- **Quantified Impact:**
  - **Baseline Recall:** ~59% (Conservative)
  - **Scheme 2 (Strict):** Gap < 2% but **Recall stuck at ~60%** (The Cost).
  - **Scheme 1/3 (Inclusion): Recall surged to 83.1%** but Husband FPR rose to 52.7%.

Recommendations for Deployment

- **1. For Social Welfare / Screening (Recommended):**
  - **Adopt Scheme 1/3 (Inclusion Model).**
  - **Logic:** The cost of missing a deserving candidate (False Negative) is higher than the cost of auditing a wrong one. Maximizes opportunity.
- **2. For High-Stakes Credit / Hiring (Alternative):**
  - **Adopt Scheme 2 (Strict Fairness).**
  - **Logic:** Legal compliance and meritocracy are paramount. We accept lower recall to ensure no "unearned advantage" is given to husbands.

Data Preparation is Governance in Action

Every imputation, transformation, and threshold choice determines who gets predicted correctly — and who doesn't. Thank you.